

THE ARMY MILITARY-CIVILIAN COST SYSTEM (AMCOS) INFLATION / DISCOUNTING AND PRESENT VALUE FACTOR (PVF)



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AMCOS INFLATION

JOINT INFLATION CALCULATOR

The Joint Inflation Calculator (JIC), as shown in Figure 1, is located on the “Query” tab in a spreadsheet that can be downloaded from the Army Financial Management & Comptroller website, <https://www.asafm.army.mil/Cost-Materials/Cost-Tools/>. In this document, the term “JIC” will refer to the above spreadsheet. It is the source for the inflation factors built into Project Manager (PM) and AMCOS Lite. See [Attachment 1](#) for further information on budget year indexes listed by AMCOS pay plan groups.. The Inflation in Project Manager and AMCOS Lite have one main difference. Project Manager only provides costs in Then Year Dollars. AMCOS Lite provides costs in constant and then year dollars. This document will discuss inflation in Project Manager primarily. The AMCOS Lite Tutorial will discuss the inflation in AMCOS Lite.

Note: All Pictures in this document can be enlarged by clicking on the Picture, holding down the control “Ctrl” Key on your keyboard, and using the scroll wheel on your mouse simultaneously.

Inflation Query Sheet CUI

1. Select Service → NAVY ARMY Marine Corps DoD Wide

ARMY

2. Select Appropriation/Cost Element from this List → Operations & Maintenance Army (OMA) Appropriation
*Note: *DON Inflation only Composite Indices are the last entries in the pick*

3. Enter Base/Output Year(1985 - 2060) 2024

Generate Inflation Table Go To SAR Calculator Worksheet

Optional - For Quick Look, complete steps A, B & C below

A. Select Inflation Type from List → Constant \$ to Then-Year \$

B. Enter Output/Target Year → 2024

C. Enter Starting Values in Input Column (blue cells) Below
Quick Look

Operations & Maintenance Army (OMA) Appropriation

4/12/2024 Print Quick Look

	Input	Inflation Factor	Output/ Result
Years	2024	→	2024
Inflation Type	Constant\$	→	TY\$
Enter starting values in the blue input cells. The inflation factor (based on your selected appropriation, year, and type) is applied and the results given in the output column	29.0	1.01874689	29.5
		1.0187	-
		1.0187	-
		1.0187	-
		1.0187	-
		1.0187	-

Figure 1: Joint Inflation Calculator

In Figure 1 above, AMCOS applies the following JIC Calculator settings before selecting the individual Appropriations (APPNs):

1. Step 1 – Select Service = “ARMY” or “DoD” are used for AMCOS.

Note: the banner shown between Steps 1 & 2 displays your service or DoD selection. In this case, the banner displays GREEN for “ARMY” and will allow ARMY APPN selections in Step 2.

2. Step 2 – Select APPN = “Military Pay Army (MPA) Appropriation (Escalation)”.

3. Step 3 – Enter Base Year/Input Year = (e.g. use the current year).
4. Step 3A – Select Inflation Type = “Then-Year \$ to Then-Year \$”.
5. Step 3B – Enter Output/Target Year = “2060”. Currently, the maximum year available is 2060.
6. Generate Inflation Table – Click on the Generate Inflation Table button. Based on the APPN selected in Step 2 for Civilians, Active Duty, NG, Reserves, and Contractor Cost Estimate, an EXCEL Inflation Table will be displayed. Use Base Year through Output/Target Year values for AMCOS tables.

In addition, AMCOS applies the “MPA Non-Pay” inflation factor only to the Active component’s PCS Costs. The “MPA Non-Pay” inflation factor may sometimes also be applied to Clothing and Allowances, but this would require a user’s adjustment. It is more important for PCS to use the MPA Non-Pay rate than Clothing & Allowances since PCS is a much larger portion of Army money.

The following picture summarizes the AMCOS inflation factors applied to Civilians, Active Duty, NG, Reserves, and Contractor Cost Estimate components. Each component has its own set of inflation rates.

CIVILIAN				ACTIVE DUTY					NATIONAL GUARD		RESERVE		Contractor Cost Estimate
CivPay	OMA-CIV	OMDW	FED-OM	MPA	MPA NonPay	OMA-MIL	OMDW	FED-OM	NGPA	OMNG	RPA	OMAR	OMA-CCE

Figure 2: Summary of AMCOS inflation factors applied to various cost estimate components

-  **Note:** The former OSD CAPE DODI cost elements were added to AMCOS as “DoD OMA” funded cost elements in Dec 2013. As part of the 15 May 2014 AMCOS release, the “DoD OMA” appropriation was separated into “DoD” & “Federal” funding levels. Presently, JIC inflation rates for the “Federal OMA” appropriation do not exist in the JIC. Until a “Federal OMA” rate is implemented, AMCOS will apply the “DoD OMA” rates to inflate AMCOS cost elements when the APPN is either “DoD OMA” or “Federal OMA”. For CONTRACTOR COST ESTIMATE positions, only the “DoD OMA” factor is applied since most contracts are paid with “OMA” dollars.
-  **Note:** The “Average Cost of Special Pay” cost element is not inflated in Project Manager.

Refer to Attachment 2 to see a list of Inflation Factors applied to DEFAULT Summary Cost Elements.

For Project Manager Purposes, we will refer to Fiscal Year (FY) costs generated in AMCOS LITE as “BASE YEAR” costs. PM multiplies BASE YEAR costs by the appropriate target years INFLATION FACTOR to generate your Future Start Year and/or your Future Year Cost Element costs across the entire duration of a project. For example, let us say you set the Project Start Year = 2023, set the Duration = 5 years, and are costing an NG position with a focus on the “Salary” cost element:

7. 2024 NG Salary = (2023 BASE YEAR Salary) * (2024 MPA-NG Inflation Factor).
8. 2025 NG Salary = (2023 BASE YEAR Salary) * (2025 MPA-NG Inflation Factor).
9. 2026 NG Salary = (2023 BASE YEAR Salary) * (2025 MPA-NG Inflation Factor).
10. 2027 NG Salary = (2023 BASE YEAR Salary) * (2027 MPA-NG Inflation Factor).
11. 2028 NG Salary = (2023 BASE YEAR Salary) * (2027 MPA-NG Inflation Factor).
- 12.
13. The Inflation Process repeats until the specified duration completes, in this case, 5 years.

ATTACHMENT 1 - JIC INFLATION CALCULATOR FACTORS USED IN AMCOS

GENERATING AMCOS INFLATION FACTORS

Inflation factors in AMCOS were generated using the below steps:

1. Use the “Query” tab of the JIC (see the Joint Inflation Calculator section).
2. Set the Service to “Army”.
3. Set the Base/Input Year to the current year.
4. Set Inflation Type to “Then-Year \$ to Then-Year \$”
5. Set Output/Target Year to “2060”
6. Set Appropriation/Cost Element to each of the below and click the **Generate Inflation Table** button.
 - a. Army-Civilian Pay (CIVPAY)
 - b. Army-Military Pay (MPA) Appropriation (Escalation)
 - c. Army-Military Pay Army National guard (MPA-NG) Appropriation Composite
 - d. Army-Military Pay Army Reserve (MPAR) Appropriation Composite
 - e. Army-Military Pay Army (MPA) Appropriation Non-Pay
 - f. Army-Operations & Maintenance Army (OMA)
 - g. Army-Operations & Maintenance Army National Guard (OMANG)
 - h. Army-Operations & Maintenance Army Reserve (OMAR)
 - i. DOD-Defense Wide O&M
 - j. DOD-Defense Wide O&M -> **NOTE THIS IS RENAMED TO FED_OM.** It is a placeholder so that in the future if one is developed for FED we can just start using that
7. Use the **Budget Year Index** values to build the tables below

ATTACHMENT 2 – DEFAULT SUMMARY INFLATION FACTOR TO COST ELEMENT MAPPINGS

APPN	JIC Mapping	CostElementCategory	CostElementName
Federal OM	FED OMA	Other Benefits	Child Education (Impact Aid)
Federal OM	FED OMA	Retirement	Treasury Contribution for Concurrent Receipts
Federal OM	FED OMA	Retirement	Treasury Contribution to MERHCF
Federal OM	FED OMA	Veteran Benefits	Veterans' Benefits (Cash and In-kind)
MPA	Army MPA	Allowances	Avg Cost of Family Separation Pay
MPA	Army MPA	Allowances	Avg Cost of Overseas Station Allowance
MPA	Army MPA	Military Compensation	Avg Cost of Base Pay (Military)
MPA	Army MPA	Military Compensation	Avg Cost of Basic Allowance for Housing
MPA	Army MPA	Military Compensation	Avg Cost of Basic Allowance for Subsistence
MPA	Army MPA	Officer Acquisition Costs	Avg Cost of Officer Acquisition
MPA	Army MPA	Other Benefits	Avg Cost of Other Benefits
MPA	Army MPA	Recruiting Costs	Avg Recruiting Cost
MPA	Army MPA	Retired Pay Accrual	Avg Cost of Retired Pay Accrual
MPA	Army MPA	Retirement	Medicare-Eligible Retiree Health Care (MERHC)
MPA	Army MPA	Selective Retention Bonus	Avg Cost of Selective Retention Bonus
MPA	Army MPA	Separation Costs	Avg Cost of All Separation Incentives
MPA	Not Inflated	Special Pays	Avg Cost of Special Pays
MPA	Army MPA	Training Costs	Avg Cost of Training
MPA Non-Pay	Army MPA Non-Pay	Permanent Change of Station Costs	Avg Permanent Change of Station-Annualized
NGPA	NG-PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)
NGPA	NG-PA	Military Compensation	Avg Cost of Basic Allowance for Subsistence
NGPA	NG-PA	Military Compensation	Avg Cost of Non-Locality Basic Allowance for Housing
NGPA	NG-PA	Other Benefits	Avg Annualized Cost of FICA
NGPA	NG-PA	Other Benefits	Avg Cost of Clothing Allowance
NGPA	NG-PA	Other Benefits	Avg Fixed Cost of GI Bill
NGPA	NG-PA	Other Benefits	Avg Fixed Cost of Student Loan Repayment Program
NGPA	NG-PA	Other Benefits	Disability, Hospitalization & Death Gratuities
NGPA	NG-PA	Recruiting Costs	Avg Recruiting Cost
NGPA	NG-PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual
NGPA	NG-PA	Selective Retention Bonus	Avg Cost of Selective Retention Bonus
NGPA	Not Inflated	Special Pays	Avg Cost of Special Pays
NGPA	NG-PA	Training Costs	Avg Cost of Training
OMA	Army OMA	Morale, Welfare and Recreation Costs	Avg Cost of Morale, Welfare and Recreation

APPN	JIC Mapping	CostElementCategory	CostElementName
OMA	Army OMA	Officer Acquisition Costs	Avg Cost of Officer Acquisition
OMA	Army OMA	Recruiting Costs	Avg Recruiting Cost
OMA	Army OMA	Training Costs	Avg Cost of Training
OMA_1	Army OMA	Officer Acquisition Costs	Avg Cost of Officer Acquisition
OMA_1	Army OMA	Training Costs	Avg Cost of Training
OMAR	Army OMA	Medical Support Costs	Avg Annualized Cost of Health Care
OMAR	Army OMA	Other Benefits	Avg Annualized Cost of Health Care
OMAR	Army OMA	Recruiting Costs	Avg Recruiting Cost
OMAR	Army OMA	Training Costs	Avg Cost of Training
OMDW	DoD OMA	Medical Support Costs	Avg Cost of Medical Support Cost
OMDW	DoD OMA	Other Benefits	Discount Groceries
OMDW	DoD OMA	Other Benefits	DoDEA and Family Assistance
OMNG	NG-OM	Medical Support Costs	Avg Annualized Cost of Health Care
OMNG	NG-OM	Other Benefits	Avg Annualized Cost of Health Care
OMNG	NG-OM	Recruiting Costs	Avg Recruiting Cost
OMNG	NG-OM	Training Costs	Avg Cost of Training
RPA	RE-PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)
RPA	RE-PA	Military Compensation	Avg Cost of Basic Allowance for Subsistence
RPA	RE-PA	Military Compensation	Avg Cost of Non-Locality Basic Allowance for Housing
RPA	RE-PA	Other Benefits	Avg Annualized Cost of FICA
RPA	RE-PA	Other Benefits	Avg Cost of Clothing Allowance
RPA	RE-PA	Other Benefits	Avg Fixed Cost of GI Bill
RPA	RE-PA	Other Benefits	Avg Fixed Cost of Student Loan Repayment Program
RPA	RE-PA	Other Benefits	Disability, Hospitalization & Death Gratuities
RPA	RE-PA	Recruiting Costs	Avg Recruiting Cost
RPA	RE-PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual
RPA	RE-PA	Selective Retention Bonus	Avg Cost of Selective Retention Bonus
RPA	Not Inflated	Special Pays	Avg Cost of Special Pays
RPA	RE-PA	Training Costs	Avg Cost of Training

CIVILIAN PAY PLANS

CIVILIAN (INTELLIGENCE PERSONNEL (GG), CIVILIAN LAW ENFORCEMENT OFFICER (GL), GENERAL SCHEDULE (GS))

APPN	JIC Mapping	CostElementCategory	CostElementName
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Benefits
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Former Employee Compensation
Army CivPay	Army CIVPAY	Cash Awards	Avg Cost of Cash Awards
Army CivPay	Army CIVPAY	Compensation - Basic	Avg Cost of Base Pay (Civilian)
Army CivPay	Army CIVPAY	Compensation - Other	Avg Cost of Other Compensation
Army CivPay	Army CIVPAY	Holiday Pay	Avg Cost of Holiday Pay
Army CivPay	Army CIVPAY	Overtime Pay	Avg Cost of Overtime Pay
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Health Insurance
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Life Insurance
OMA	Army OMA	Training Costs	Training

SENIOR EXECUTIVE SERVICE SCHEDULE (SES)

APPN	JIC Mapping	CostElementCategory	CostElementName
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Army-Funded Retirement
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Federal Employees Gov't Health Insurance
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Federal Employees Gov't Life Insurance
Army CivPay	Army CIVPAY	Benefits	Avg Cost of FICA
Army CivPay	Army CIVPAY	Cash Awards	Avg Cost of Cash Awards
Army CivPay	Army CIVPAY	Compensation - Basic	Avg Cost of Base Pay (Civilian)
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Health Insurance
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Life Insurance
OMA	Army OMA	Training Costs	Training

GFEBs BASED (DB, DE, DJ, DK, NH, NJ, NK,GP)

APPN	JIC Mapping	CostElementCategory	CostElementName
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Army-Funded Retirement
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Federal Employees Gov't Health Insurance
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Federal Employees Gov't Life Insurance
Army CivPay	Army CIVPAY	Benefits	Avg Cost of FICA

APPN	JIC Mapping	CostElementCategory	CostElementName
Army CivPay	Army CIVPAY	Benefits	Civ Employer Share FEGLI (Fed Employees Group Life Insurance) (6400.12K0)
Army CivPay	Army CIVPAY	Benefits	Civ Employer Share FEHB (Fed Employees Group Health Benefit Insurance) (6400.12N0)
Army CivPay	Army CIVPAY	Benefits	Civ Employer Share FICA / Medicare (6400.12Q0)
Army CivPay	Army CIVPAY	Benefits	Civ Employer Share Retirement (6100.12Y0, 6400.12L0, 6400.12M0, 6400.12X0)
Army CivPay	Army CIVPAY	Benefits	Civ Non-Foreign COLA (Cost of Living Allowance) Pay (6100.12C0)
Army CivPay	Army CIVPAY	Benefits	Civ Overseas Allowances (Civ Quarters, COLA, LQA, & Other not classified) (6100.12B0)
Army CivPay	Army CIVPAY	Cash Awards	Avg Cost of Cash Awards
Army CivPay	Army CIVPAY	Cash Awards	Civ Cash Awards Pay (6100.11K0)
Army CivPay	Army CIVPAY	Compensation - Basic	Avg Cost of Base Pay (Civilian)
Army CivPay	Army CIVPAY	Compensation - Basic	Civ Base Pay (6100.11B1)
Army CivPay	Army CIVPAY	Compensation - Basic	Civ Locality Pay
Army CivPay	Army CIVPAY	Compensation - Basic	Civ Title 38: Medical Premium Pay (6100.11N0)
Army CivPay	Army CIVPAY	Compensation - Other	Civ Hazardous Duty Pay (6100.11H0)
Army CivPay	Army CIVPAY	Compensation - Other	Civ Physician Comparability Pay (Market Pay) (6100.11T0)
Army CivPay	Army CIVPAY	Compensation - Other	Civ Post Differential Pay (O/S Hardship Post) (6100.11J0)
Army CivPay	Army CIVPAY	Compensation - Other	Civ Supervisory Special Pay (6100.11Q0)
Army CivPay	Army CIVPAY	Other	Civ Other Benefits not Otherwise Classified (6100.12S2)
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Health Insurance
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Life Insurance
OMA	Army OMA	Training Costs	Training

CIVILIAN WAGE PAY PLANS

W, X SERIES & NAF (NA,NS,NL)

APPN	JIC Mapping	CostElementCategory	CostElementName
Army CivPay	Army CIVPAY	Benefits	Avg Annualized Cost of FICA
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Army-Funded Retirement
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Federal Employees Gov't Health Insurance
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Federal Employees Gov't Life Insurance
Army CivPay	Army CIVPAY	Benefits	Avg Cost of Miscellaneous Pay
Army CivPay	Army CIVPAY	Cash Awards	Avg Cost of Cash Awards
Army CivPay	Army CIVPAY	Compensation - Basic	Avg Cost of Base Pay (Civilian)
Army CivPay	Army CIVPAY	Compensation - Other	Avg Cost of Premium Pay
Army CivPay	Army CIVPAY	Compensation - Other	Avg Cost of Overtime Pay

APPN	JIC Mapping	CostElementCategory	CostElementName
Army CivPay	Army CIVPAY	Former Employee Comp	Avg Cost of Former Employee Compensation
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Health Insurance
Federal OM	FED OMA	Retired Pay Accrual	Avg Cost of Post Retirement Life Insurance
OMA	Army OMA	Training Costs	Training

CONTRACTOR COST ESTIMATE (CCE)

APPN	JIC Mapping	CostElementCategory	CostElementName
CONTRACTOR	Army OMA	Compensation	Avg Cost of Salary
CONTRACTOR	Army OMA	Compensation	Avg Cost of Benefits
CONTRACTOR	Army OMA	Overhead	Overhead

AMCOS DISCOUNTING AND PRESENT VALUE FACTOR (PVF)**DISCOUNTING AND PRESENT VALUE INFORMATION**

Most cost comparison techniques take into consideration the time value of money, that is, a dollar today is worth more than the same dollar in the future except in times of negative inflation rates. There are three reasons this is true. Opportunity cost: Money available today can be invested and accrue interest increasing its value. Inflation: Money may buy less in the future than it does today. Uncertainty: Something could happen to the money before a person is scheduled to receive it. Discounting is the process of converting a value received in a future time period to an equivalent value received immediately. For comparison purposes, future expenditures, occurring at different points in time, must be adjusted to a common point in time. Discount factors can be calculated once an interest rate and period of analysis are determined. These discount factors indicate the present value (today's value) of a dollar expended at mid-year or the end of each respective year (AMCOS applies mid-year discounting).

Office of Management and Budget (OMB) Circular A-94 and Department of Defense Instruction (DoDI) 7041.04 require the use of a Discount Rate based on the Treasury Department cost of borrowing funds. This Discount Rate should be used in evaluating the measurable costs and/or benefits of programs or projects when they are distributed over time. The prescribed rate will vary dependent on the length of the period of analysis and on whether the costs and benefits are measured in constant or current dollars. A Discount Rate that reflects expected inflation should be used to discount costs and benefits expressed in current dollars, as is the case in AMCOS. Conversely, a Discount Rate that has already been adjusted to eliminate the effect of expected inflation should be used to discount costs and benefits expressed in constant dollars.

The Discount Rate for use in DoD economic analyses is based on the U.S. Treasury Department's cost of borrowing funds and reflects the expected cost of borrowing for 3, 5, 7, 10, 20, and 30-year securities. Annually, the OMB issues an update of the borrowing rates based on the estimates provided in the President's budget submission. Typically, the update is issued early in the calendar year and remains applicable until the next annual update.

AMCOS uses the six basic rates (3, 5, 7, 10, 20, and 30-year securities) and the user's Project Manager Duration input determines which rate applies.

AMCOS DISCOUNTING AND PRESENT VALUE REVIEW

During April 2021, DASA-CE reviewed the AMCOS Discounting methodology and concluded the following:

- 14. Project Manager uses then-year dollars that include expected inflation**
- 15. Nominal discount rates should be used for expected inflation or future purchasing power.**
- 16. FINDING: Apply the nominal Discount Rates for AMCOS.**

DISCOUNTING AND PRESENT VALUE GUIDANCE

17. DoD INSTRUCTION Number 7041.04 (Link: <http://www.esd.whs.mil/Directives/issuances/dodi/>)
October 2, 2017

SUBJECT: Economic Analysis for Decision Making

REF: Page 82

Treatment of Inflation. All budget estimates must be in current (also known as “Then-Year”) dollars. For analytical purposes, all estimates of the costs and benefits for each year of the period of analysis can be made in either of two ways:

- Constant dollars that measure costs and benefits for stable purchasing power; or
- Current dollars that measure costs and benefits for future purchasing power of the dollar. In a single analysis, computations should not mix current and constant dollars

REF: Appendix 3 to Enclosure 2. DETERMINING THE DISCOUNT RATE FOR PERFORMING ECONOMIC

ANALYSIS OF INVESTMENT ALTERNATIVES

When costs and benefits occur in a steady stream, applying mid-year factors is best for the analysis. The formula for the calculation of the mid-year discount factors is

$$F_n = 1/(1+i)^{(n-.5)}$$

18. Department of the Army
Economic Analysis Manual
U.S. Army Cost and Economic Analysis Center
<https://apps.dtic.mil/dtic/tr/fulltext/u2/a437381.pdf>
May 2002

REF: Page 35

OMB Circular A-94 and DoDI 7041.04 require the use of a Discount Rate based on the Treasury Department cost of borrowing funds. This Discount Rate should be used in evaluating the measurable costs and benefits of programs or projects when they are distributed over time. The prescribed rate will vary dependent on the length of the period of analysis and on whether the costs and benefits are measured in constant or current dollars. A Discount Rate that has already been adjusted to eliminate the effect of expected inflation should be used to discount costs and benefits expressed in constant dollars. Conversely, a Discount Rate that reflects expected inflation should be used to discount costs and benefits expressed in current dollars.

19. OMB Circular No. A-94
Discount Rates for Cost-Effectiveness
<https://www.whitehouse.gov/omb/information-for-agencies/circulars/>

Nominal Discount Rates. A forecast of nominal or market interest rates based on economic assumptions. These nominal rates are to be used for discounting nominal flows, which are often encountered in lease-purchase analysis.

AMCOS PROJECT MANAGER DISCOUNTING AND PRESENT VALUE FACTOR (PVF):

Project Manager automatically calculates the Discount Rate using the Present Value Factor (PVF) calculation, which will be discussed shortly. You may be thinking, “we’re costing people (military or civilian); it’s not as though we’re costing buildings; do we really need to discount?” The answer is yes, the money spent on people still follows the same principles it follows when being used for buildings. Money today is always more preferable than money received at a later time. What about projects that are only 1-2 years in duration? Again, the answer is yes because there is still going to be a reduction in value in the money one or two years from now versus what the money is worth today.

There is no need to turn discounting on and off in Project Manager. Discounting is automatically calculated based on the duration of the project and displays in the Project Manager’s output and exportable report. Output products provide both cost scenarios, Undiscounted (**without** discounting) and Discounted (**with** discounting), as well as displays a Discount Rates recap table applied by the model.

DASA-CE approved the following business rules for incorporation into the model:

1. **Duration of the project will determine which rate to use.**
2. **Use Mid-Year Discount Factors.** The mid-year exponent is expressed in the .5 power and will start during the first year with the 0.5 Exponent for the BASE YEAR. Example: 0.5, 1.5, 2.5, 3.5, 4.5, 5.5, etc.
3. **Use the latest Nominal Rates cited by the annual DoD Memo regarding Discount Rates for OMB Circular No. A-94.**
4. **Calculate PVF values based on the PVF formula.**
5. **To get the Discounted Cost, multiply the (PM Reports Undiscounted Cost Elements) by (PVF factor).**

THE PRESENT VALUE FACTOR (PVF) CALCULATION:

20. **Determine which securities rate to use, 3-Year, 5-Year, etc.:** The model applies a separate Discount Rate for each year across the "Duration" of the project. The project’s duration will determine which rate to use. For example, the user sets the Project’s Duration = 6 years which falls in the 6 but less than 8 years range, therefore, the 7-Year Rate nominal interest rate will be applied.

YEAR	LESS THAN	RATE Applied
1	4	3-Year Rate
4	6	5-Year Rate
6	8	7-Year Rate
8	11	10-Year Rate
11	21	20-Year Rate
21 or >	-	30-Year Rate

Use Mid-Year Discount Factors: **For this 7-Year Rate example, the exponent, “n” will equal:**

Year 1 = 0.5

Year 2 = 1.5

Year 3 = 2.5

Year 4 = 3.5

Year 5 = 4.5

Year 6 = 5.5

Year 7 is NOT REQUIRED for a six-year project; AMCOS just requires the correct nominal interest rate.

- 21. Determine the Nominal Interest Rate:** Use the example below to determine the rate to use for your analysis. <https://www.whitehouse.gov/wp-content/uploads/2023/12/CircularA-94AppendixC.pdf>

**Nominal Interest Rates on Treasury Notes and Bonds
of Specified Maturities (in percent)**

<u>3-Year</u>	<u>5-Year</u>	<u>7-Year</u>	<u>10-Year</u>	<u>20-Year</u>	<u>30-Year</u>
4.5	4.4	4.4	4.4	4.7	4.7

- 22. Calculate PVF Values:** The methodology for calculating the discount factors associated with the various Discount Rates is based on the following formula:

$$PVF_n = 1/(1+i)^{(n-.5)}$$

Where PVF = the present value factor for year “n”

i = the Discount Rate

n = the project year

Continuing with our 7-Year Rate example, the calculation of the mid-year discount factors for a six-year project using current dollars is:

PV	Formula	Factor
PVF ₁	$1/(1+0.044)^{(0.5)}$	0.97870
PVF ₂	$1/(1+0.044)^{(1.5)}$	0.93745
PVF ₃	$1/(1+0.044)^{(2.5)}$	0.89794
PVF ₄	$1/(1+0.044)^{(3.5)}$	0.86010
PVF ₅	$1/(1+0.044)^{(4.5)}$	0.82385
PVF ₆	$1/(1+0.044)^{(5.5)}$	0.78913
PVF ₇ is not required for a six-year project		

23.

Figure 2 shows the PVF for all six Nominal Discount Rates.

Year	Mid-Year Exponent	3 Year 4.5	5 Year 4.4	7 Year 4.4	10 Year 4.4	20 Year 4.7	30 Year 4.7
1	0.5	0.97823	0.97870	0.97870	0.97870	0.97730	0.97730
2	1.5	0.93611	0.93745	0.93745	0.93745	0.93343	0.93343
3	2.5	0.89580	0.89794	0.89794	0.89794	0.89152	0.89152
4	3.5		0.86010	0.86010	0.86010	0.85150	0.85150
5	4.5		0.82385	0.82385	0.82385	0.81328	0.81328
6	5.5			0.78913	0.78913	0.77677	0.77677
7	6.5			0.75587	0.75587	0.74190	0.74190
8	7.5				0.72401	0.70860	0.70860
9	8.5				0.69350	0.67679	0.67679
10	9.5				0.66427	0.64641	0.64641
11	10.5					0.61739	0.61739
12	11.5					0.58968	0.58968
13	12.5					0.56320	0.56320
14	13.5					0.53792	0.53792
15	14.5					0.51378	0.51378
16	15.5					0.49071	0.49071
17	16.5					0.46868	0.46868
18	17.5					0.44764	0.44764
19	18.5					0.42755	0.42755
20	19.5					0.40836	0.40836
21	20.5						0.39003
22	21.5						0.37252
23	22.5						0.35579
24	23.5						0.33982
25	24.5						0.32457
26	25.5						0.31000
27	26.5						0.29608
28	27.5						0.28279
29	28.5						0.27010
30	29.5						0.25797

Figure 2: PVF for all six Nominal Discount Rates

24. **Calculate the Discounted Cost:** Multiply the (PM Reports Undiscounted Cost Elements) by (PVF factor). Finalizing our 7-Year Rate example, the calculation of the mid-year discount factors for a six-year project using current dollars is:

PV	Formula	Factor	Undiscounted Cost	Discounted Cost
PVF ₁	$1/(1+0.044)^{(0.5)}$	0.97870	\$ 1,000.00	\$ 978.70
PVF ₂	$1/(1+0.044)^{(1.5)}$	0.93745	\$ 1,000.00	\$ 937.45
PVF ₃	$1/(1+0.044)^{(2.5)}$	0.89794	\$ 1,000.00	\$ 897.94
PVF ₄	$1/(1+0.044)^{(3.5)}$	0.86010	\$ 1,000.00	\$ 860.10
PVF ₅	$1/(1+0.044)^{(4.5)}$	0.82385	\$ 1,000.00	\$ 823.85
PVF ₆	$1/(1+0.044)^{(5.5)}$	0.78913	\$ 1,000.00	\$ 789.13
PVF ₇ is not required for a six-year project				